Visual Analytics and Network Analysis of Facebook Engagement Using Python

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Abstract

An analysis is presented of 7,050 anonymized Facebook posts, and seven reproducible network-visualization exercises are completed. Cleaning, feature engineering, static and interactive visualization, graph statistics, testing, and public web delivery are combined. The largest observed mean total engagement is associated with video posts (1,041.57), but causality is not established by the observational design.

Keywords: Facebook engagement, data visualization, social-network analysis, NetworkX, Plotly, Python.

1. Introduction and research questions

Multidimensional engagement is described while the distinction between real tabular observations and synthetic relationship graphs is preserved. Variation by content and time, structural readability under different visual layouts, and the social-network properties captured by canonical graph models are examined.

A central methodological constraint is that the Facebook table contains no verified user-to-user edges. The empirical analysis therefore remains tabular; all assignment relationship graphs are explicitly synthetic.

2. Dataset, license, and ethics

Facebook Live Sellers in Thailand was authored by Nassim Dehouche and distributed by UCI under CC BY 4.0. It records anonymized post engagement from 2012-07-15 through 2018-06-13.

The official UCI archive was used because Kaggle credentials were not configured. The Kaggle mirror is documented but was not the acquisition route. No names, message text, or re-identification attempts are included.

3. Data cleaning and feature engineering

The raw table contained 7,050 rows and 16 columns. Four empty placeholder columns were removed; 0 duplicate rows and 0 invalid dates were found. The processed table contains 7,050 rows and 27 columns.

Total engagement equals reactions plus comments plus shares. Ratio features use zero-safe division. Temporal features include posting hour, weekday, month, ISO week, weekend, and time-of-day. No engagement rate is claimed because reach and impressions are absent.

4. Exploratory engagement findings

The data record 1,622,326 reactions, 1,581,710 comments, and 282,159 shares. Mean total engagement is 494.50, while the median is 69.00; the gap is consistent with strong right skew.

Type	Posts	Mean reactions	Mean comments	Mean shares	Mean total
Video	2,334	283.41	642.48	115.68	1,041.57
Status	365	438.78	36.24	2.56	477.58
Link	63	370.14	5.70	4.40	380.24
Photo	4,288	181.29	15.99	2.55	199.84

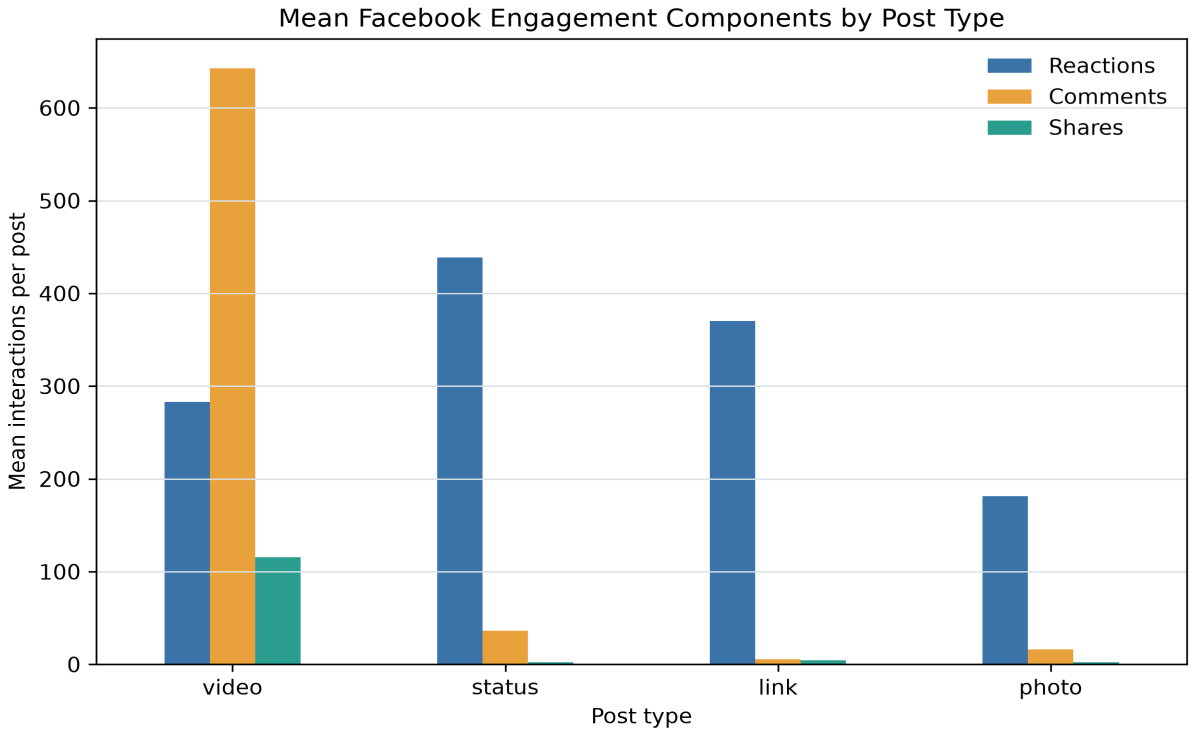

Figure 1. Mean engagement components by post type.

5. Distribution, correlation, and time

The upper Tukey fence flags 972 posts rather than deleting them. Rank correlations and log-transformed distributions are used because extreme posts make raw-count views difficult to read.

The largest observed median engagement occurs at 18:00, but content mix, seller behavior, and seasonality are plausible confounders.

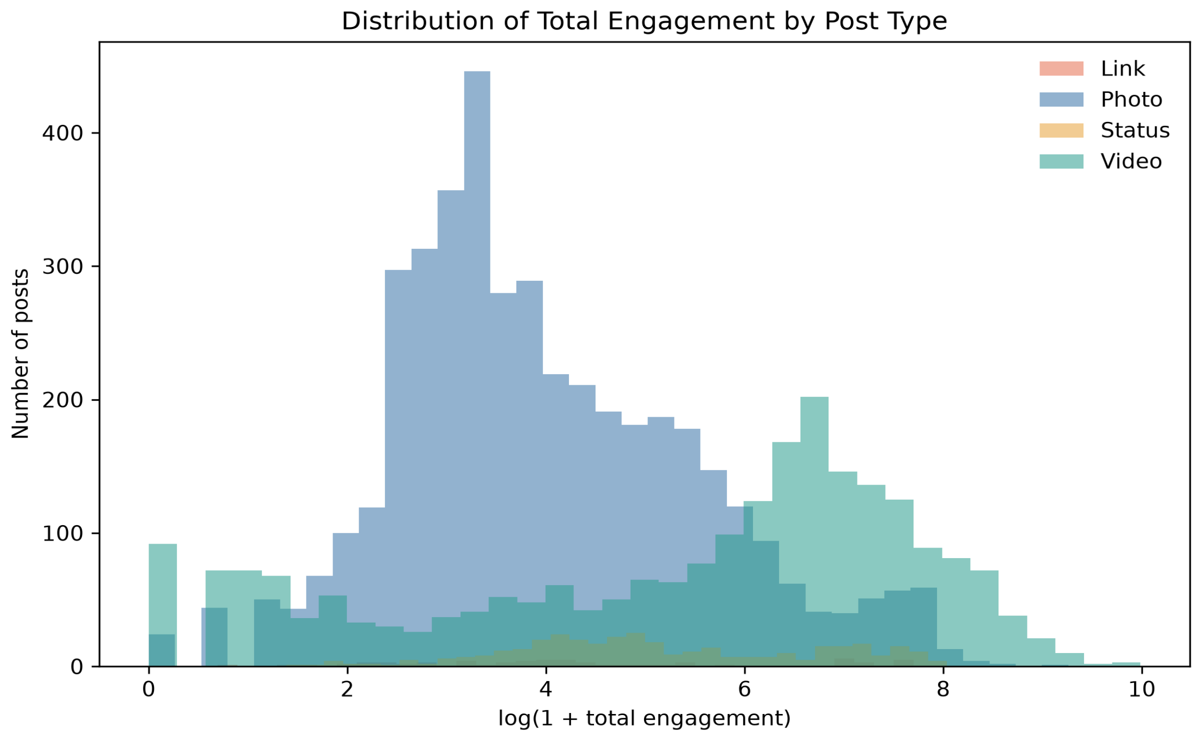

Figure 2. Log-transformed total-engagement distributions by post type.

5.1 Rank-correlation result

Spearman rank correlation was used because the engagement variables are strongly skewed. The coefficients describe monotonic association and were not interpreted as causal effects.

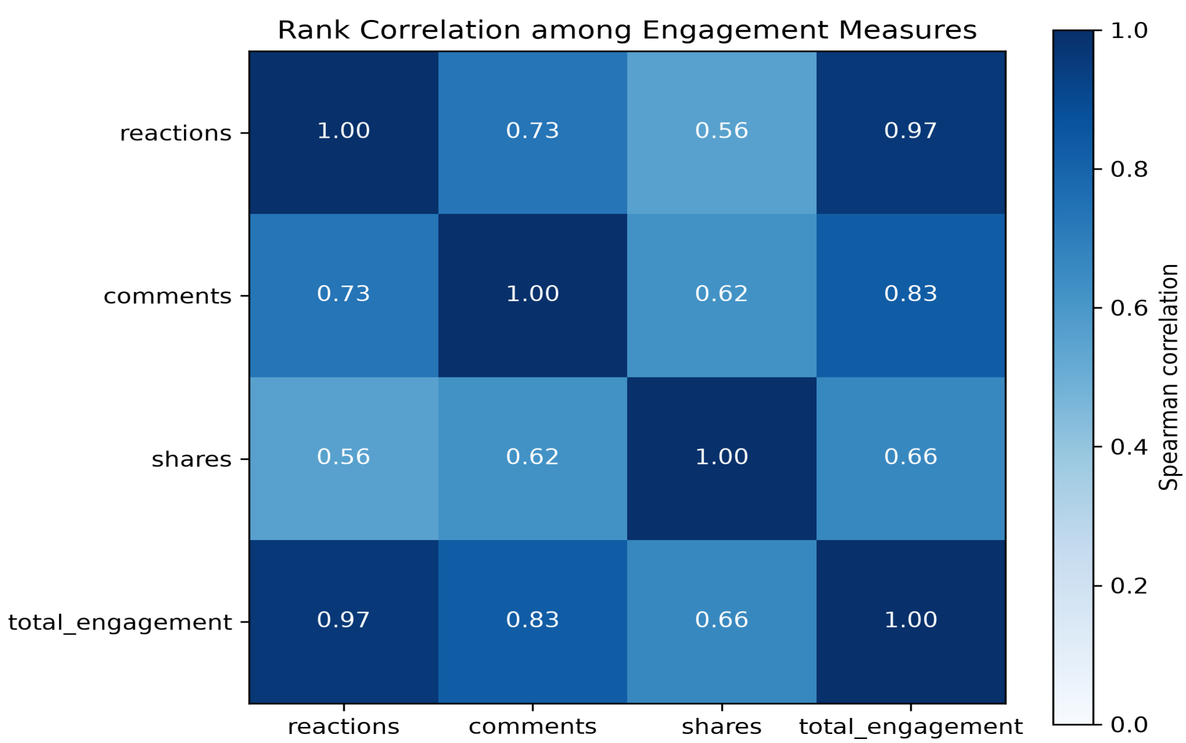

Figure 3. Spearman correlation matrix for engagement measures.

6. Graph representation and visualization methodology

Adjacency matrices support exact pairwise lookup; node-link diagrams emphasize paths, clusters, bridges, and hubs. Visual encodings are held constant during layout comparison so that topology-driven placement is the primary changing variable.

Force-directed algorithms map graph-theoretic relationships to proximity, whereas circular and random layouts do not optimize community readability.

7. Exercise 1 - Weighted adjacency representations

The relationship between an adjacency list and a weighted adjacency matrix was examined through the supplied G_transport network. Matrix symmetry and its implication for route direction were evaluated.

7.1 Objective, construction, and procedure

Seven cities and 9 undirected teaching routes were represented. Distance in kilometers was stored in every edge weight. Density was 0.4286, and Dhaka received the largest degree (6).

A labeled adjacency list was produced from neighbor iterators. A weighted matrix was generated with nx.to_pandas_adjacency, and symmetry was tested by comparison with the transpose through numpy.allclose. Off-diagonal zeros were treated as absent routes.

7.2 Verified results

The symmetry test returned True. The Chattogram-Sylhet edge was longest at 350 km, while the Khulna-Barishal edge was shortest at 130 km. Unique edge distances totaled 2,056 km.

City	Dhaka	Chattogram	Sylhet	Khulna	Rajshahi	Barishal	Rangpur
Dhaka	0	264	247	209	256	170	300
Chattogram	264	0	350	0	0	0	0
Sylhet	247	350	0	0	0	0	0
Khulna	209	0	0	0	0	130	0
Rajshahi	256	0	0	0	0	0	130
Barishal	170	0	0	130	0	0	0
Rangpur	300	0	0	0	130	0	0

7.3 Interpretation, limitations, and verification

Symmetry was produced because every route was encoded as an undirected edge with one shared weight. An asymmetric matrix would be expected under one-way travel or direction-dependent cost. The adjacency list was better suited to neighbor inspection, while the matrix supported exact pairwise lookup.

The graph was treated as a classroom illustration rather than a complete transportation model. The matrix was saved as a CSV, and symmetry is covered by an automated graph-representation test.

8. Exercise 2 - Six-layout comparison

Six layouts were compared for one unchanged G_ppi topology. Identity, labels, node size, color, and edges were fixed so that placement was the only changing visual variable.

8.1 Graph construction and controlled procedure

The graph was reproduced with nx.watts_strogatz_graph using n=15, k=3, p=0.3, and seed 42, then relabeled P1 through P15. The realization contains 15 edges, density 0.1429, average degree 2.00, diameter 8, and degree range 1-4.

Spring, circular, shell, spectral, Kamada-Kawai, and random placement were assessed through crossings, label separation, ring locality, shortcut visibility, and correspondence between graph distance and visual proximity.

8.2 Verified result and critical interpretation

Kamada-Kawai was selected for this realization because pairwise distance stress was minimized and ring-like locality remained legible. Spring placement was a close alternative.

Average clustering was 0.000. Triangle-based clustering was absent in this exact seeded graph, so the layouts were interpreted as views of locality and shortcuts rather than as evidence of communities.

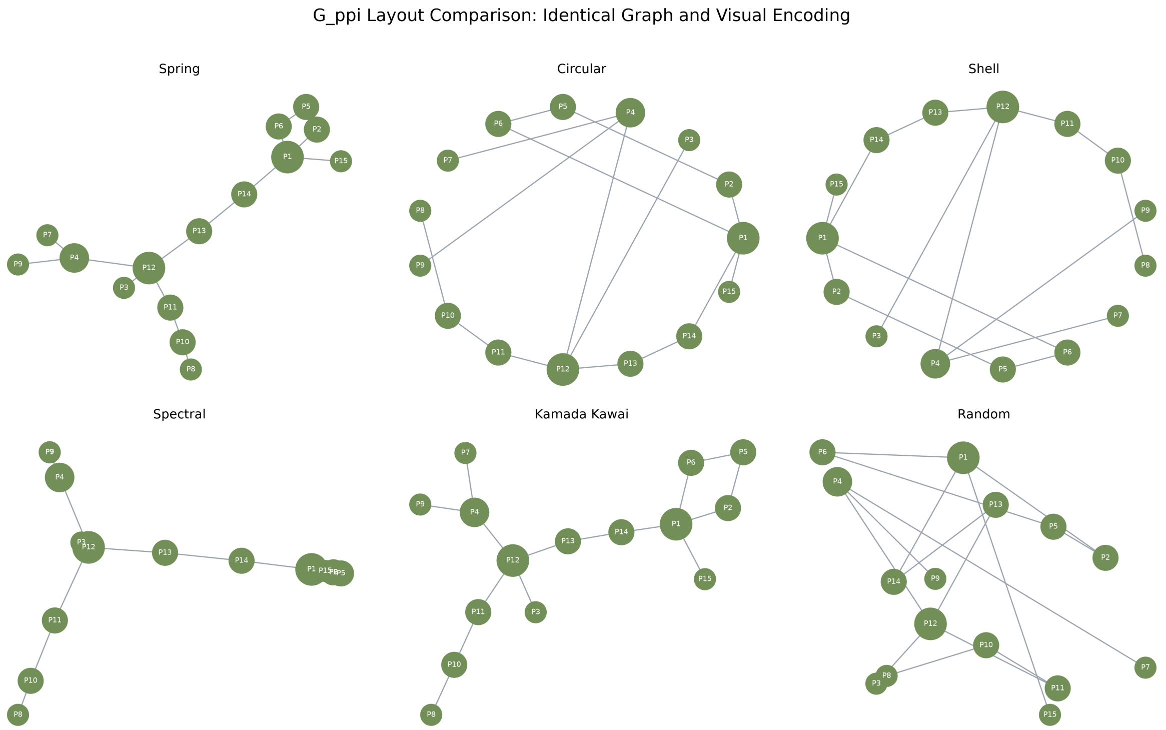

Figure 4. Six layouts of the identical synthetic G_ppi graph.

8.3 Limitations and verification

Layout preference remains dependent on graph size, parameters, task, and labeling. Six individual figures and a combined comparison were saved, and all reported graph statistics were recalculated from the generated NetworkX object.

9. Exercise 3 - Student-course bipartite network

A two-mode enrollment network was constructed without inventing student-student or course-course edges.

9.1 Data design, construction procedure, and validation

The stored synthetic data contain 12 students, 6 courses, and 40 enrollment records. Students were assigned to partition 0 and courses to partition 1.

An edge was added only when a stored enrollment linked the two partitions. Bipartiteness returned True, and bipartite density was 0.5556.

9.2 Degree results

Mean course load was 3.33 per student, and mean course enrollment was 6.67. Data Visualization had the largest course degree (12), while Imran Kabir had the largest student degree (5).

Course	Student enrollments
Data Visualization	12
Machine Learning	7
Social Network Analysis	6
Database Systems	5
Human-Computer Interaction	5
Research Methodology	5

9.3 Visual result and interpretation

Separate columns, shapes, and colors made partition membership explicit. Student degree represented course load, whereas course degree represented synthetic popularity. A one-mode projection was not used because derived ties were outside the exercise scope.

Data Visualization bridges every represented major. Data Science students also concentrate in Machine Learning and Social Network Analysis, while Multimedia Technology students favor Human-Computer Interaction.

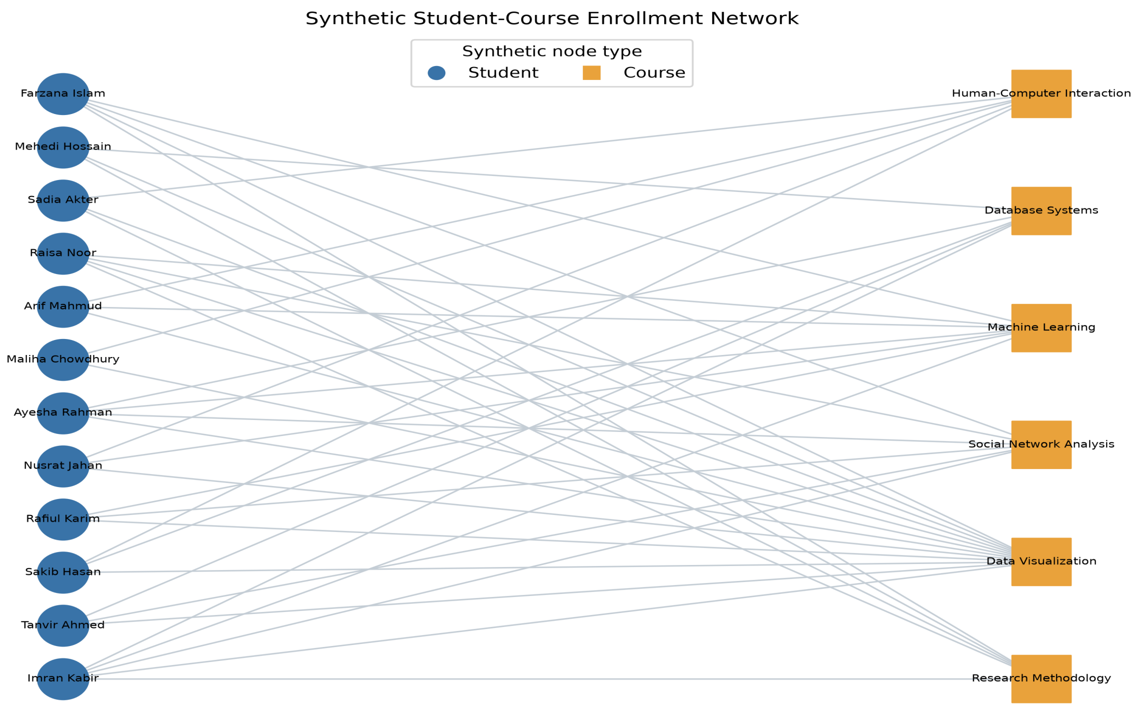

Figure 5. Synthetic student-course enrollment network.

9.4 Limitations and verification

Names and enrollment records were generated solely for teaching. No claim was made about actual students or enrollment behavior. Source tables, calculated degree results, and an automated partition test provide reproduction evidence.

10. Exercise 4 - Weighted Barabasi-Albert graph

The perceptual effect of edge-width encoding was examined while the underlying 100-node preferential-attachment topology was preserved.

10.1 Construction and weight assignment

The topology was generated with n=100, m=2, and seed 42. It contains 196 edges, average degree 3.92, and maximum degree 36.

Each copied edge received a reproducible integer weight from 1 through 10. Edge width was scaled from weight; topology, degree, and node positions were not changed.

10.2 Verified weight distribution

The observed range was 1-10, the mean was 5.388, the median was 5.0, and the population standard deviation was 2.852. Total assigned edge weight was 1,056.

Edge weight	Number of edges
1	16
2	26
3	18

Edge weight	Number of edges
4	25
5	20
6	18
7	17
8	14
9	25
10	17

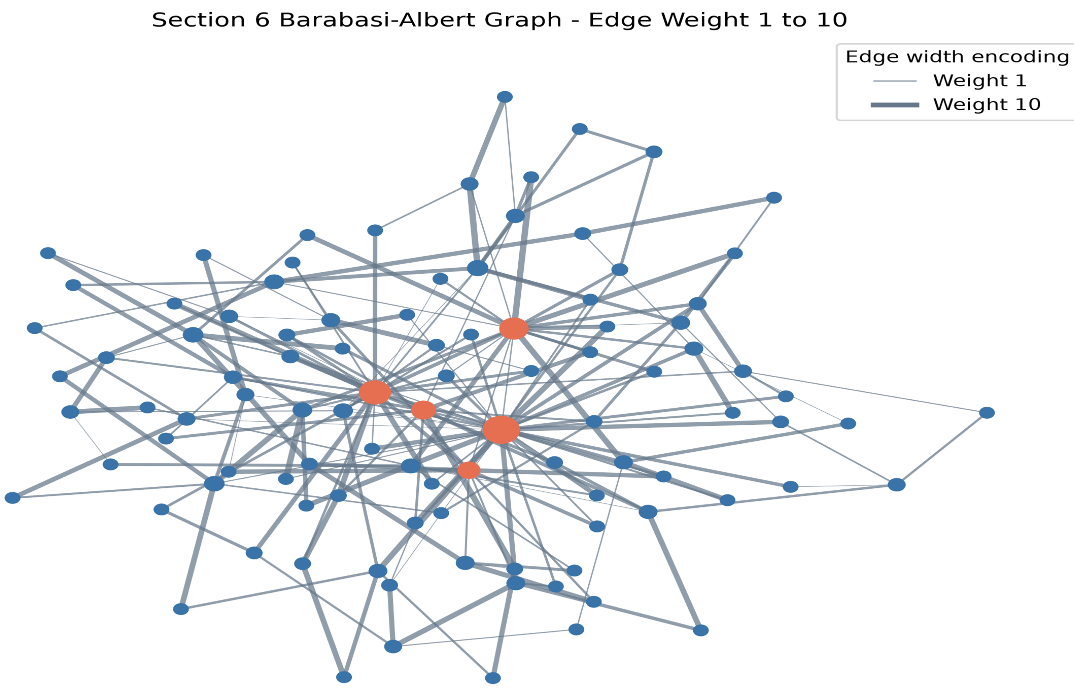

Figure 6. Barabasi-Albert graph with edge-width encoding.

10.3 Interpretation, limitations, and verification

Thicker lines increased tie salience while hub structure remained unchanged. Edge weight was not treated as node centrality. A weighted centrality analysis would require weight to be defined as strength, capacity, cost, or distance.

The random weights demonstrate encoding only. The edge table was saved, and automated tests verify graph order, size, range, and determinism.

11. Exercise 5 - Generative-model comparison

Random mixing, local clustering with rewiring, and preferential attachment were compared under the exact supplied parameters.

11.1 Model construction, parameters, and metric procedure

Erdos-Renyi G(30, 0.08), Watts-Strogatz WS(30, 4, 0.1), and Barabasi-Albert BA(30, 2) were generated with seed 42.

Order, size, density, average degree, clustering, connectivity, components, average path length, maximum degree, and degree standard deviation were calculated. For the disconnected Erdos-Renyi graph, path length was calculated on the largest connected component.

11.2 Verified statistical results

Model	Nodes	Edges	Density	Clustering	Avg. path	Components	Max degree
Erdos-Renyi G(n,p)	30	33	0.076	0.072	3.828	3	4
Watts-Strogatz small-world	30	60	0.138	0.346	2.984	1	5
Barabasi-Albert scale-free	30	56	0.129	0.300	2.182	1	18

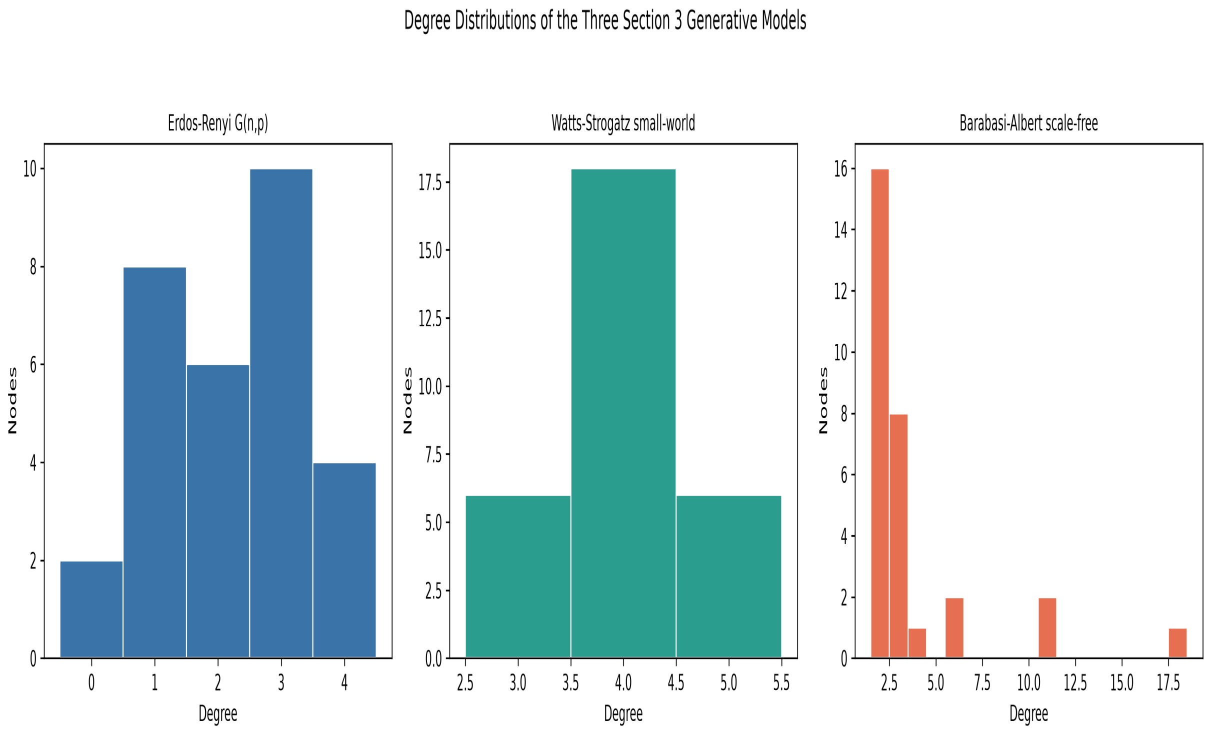

Figure 7. Degree distributions of the three seeded models.

11.3 Model-by-model interpretation

The Erdos-Renyi realization produced three components and low clustering (0.072); it served as a homogeneous random baseline.

The Watts-Strogatz realization remained connected and produced the highest clustering (0.346) with a narrow degree distribution.

The Barabasi-Albert realization remained connected, produced the shortest average path (2.182), and generated maximum degree 18 with strong degree heterogeneity.

Watts-Strogatz captures clustering and short paths; Barabasi-Albert captures hubs and heterogeneous degree. A social network can require both properties, so no model is universally best.

11.4 Limitations and verification

Only one small seeded realization was compared per model. Repeated simulation and uncertainty intervals would be required for general model inference. The comparison CSV and separate degree-distribution figures provide reproduction evidence.

12. Exercise 6 - Interactive node-color dashboard

One directed graph was interpreted through categorical interest-group color and continuous in-degree color without changing node positions.

12.1 Graph construction procedure and results

The seeded logic produced 20 users and 52 directed follow edges. Directed density was 0.1368. Mean in-degree was 2.60; in-degree ranged from 1 to 5, and out-degree ranged from 1 to 5.

The three largest in-degree results were user_11 (5), user_12 (4), user_13 (4).

Interest group	Users
Music	2
Sports	8
Tech	7
Travel	3

12.2 Initial categorical-color state

Categorical color was used to support comparison of synthetic interest groups. Node coordinates, edges, labels, and hover fields were fixed.

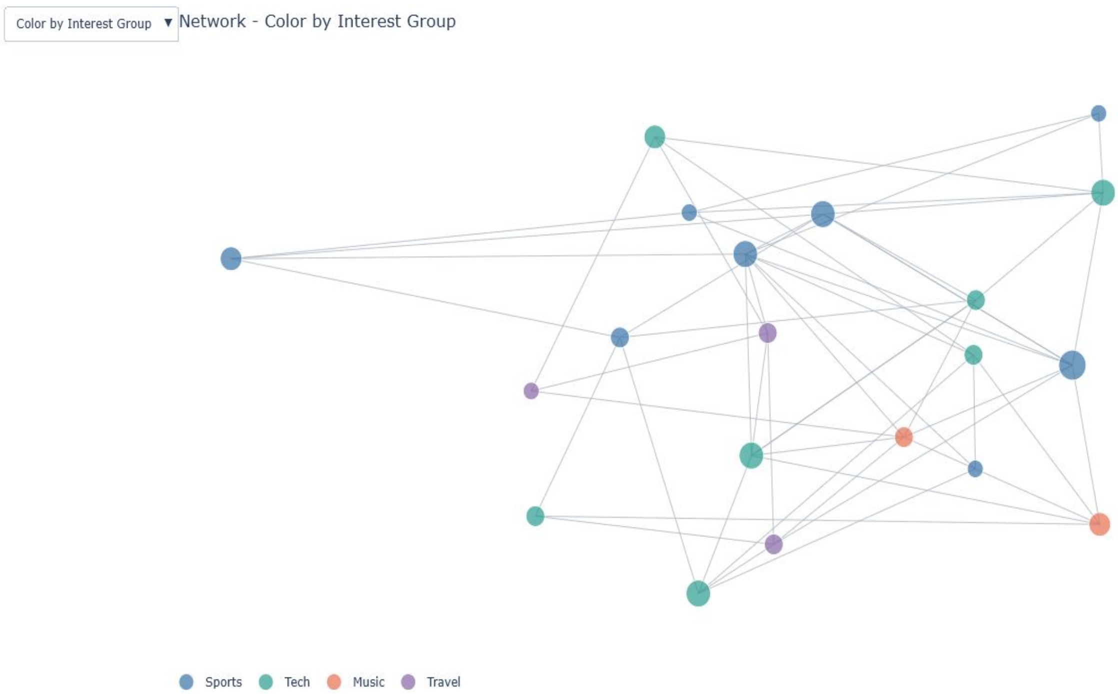

Figure 8. Interactive network colored by interest group.

12.3 Continuous in-degree state

The dropdown changed marker color, legend, and scale metadata to in-degree while coordinates remained fixed. Position preservation was verified as True.

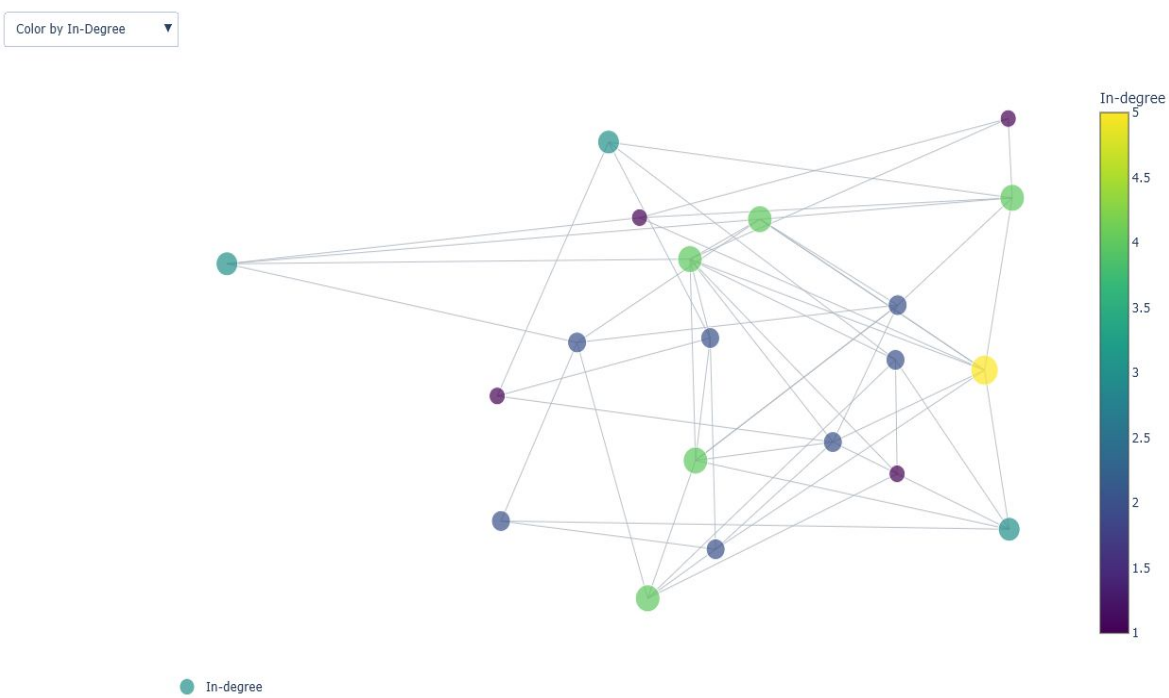

Figure 9. Interactive network colored by in-degree.

12.4 Interpretation, limitations, and verification

The categorical state supported group-mixing interpretation, while the continuous state emphasized incoming-tie popularity. Perceptual change was attributable to color because position was preserved.

The graph is synthetic, and Plotly line traces do not display arrowheads. Direction remains available through degree metrics. Both dropdown states were verified through browser interaction.

13. Exercise 7 - Research-domain knowledge graph

A typed synthetic graph was used to represent research areas, methods, tools, applications, outcomes, and semantic relationships.

13.1 Construction and metric procedure

The graph contains 15 nodes and 21 undirected links. Density was 0.2000, average degree was 2.80, and connectivity was verified as True.

Degree centrality, inverse-strength weighted betweenness, closeness, and strength-weighted PageRank were calculated. Edge distance was defined as 1 / strength before shortest-path betweenness was evaluated.

13.2 Verified centrality results

Recommendation Systems received the largest weighted betweenness value (0.5385).

Node	Type	Degree	Betweenness	Closeness	PageRank
Recommendation Systems	Research area	5	0.5385	0.5600	0.1297
Artificial Intelligence	Research area	3	0.2509	0.4667	0.0790
Deep Learning	Method	4	0.1923	0.4667	0.1011
Computer Vision	Research area	3	0.1667	0.4375	0.0795
Graph Neural Networks	Method	3	0.1612	0.5000	0.0630

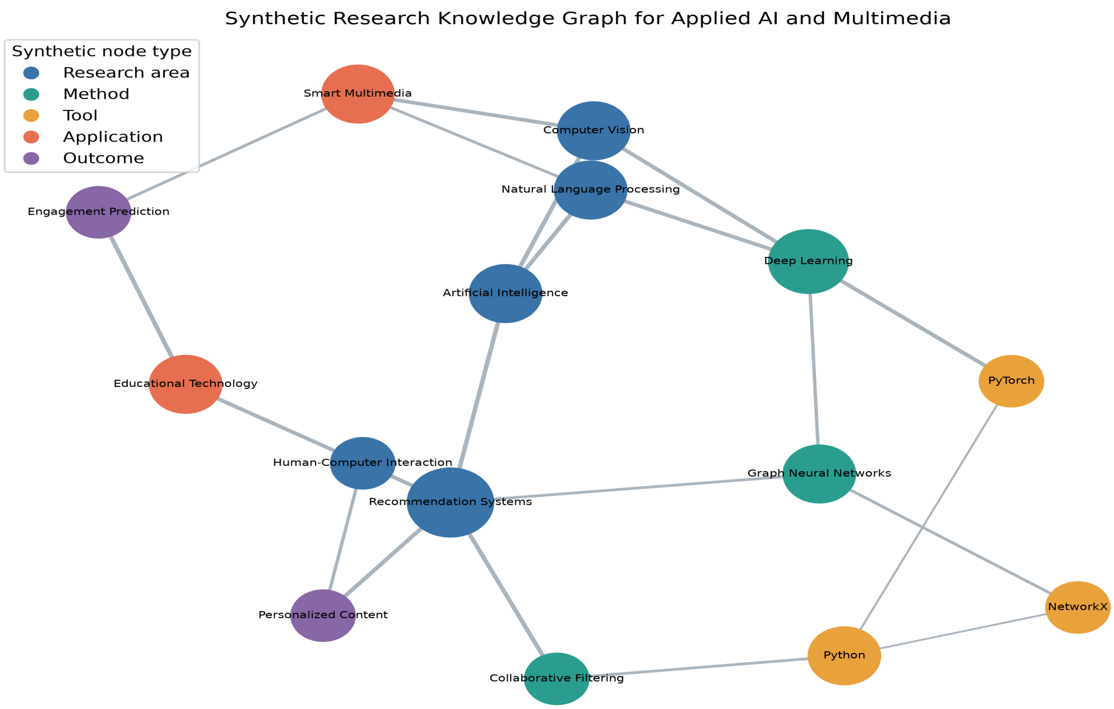

Figure 10. Synthetic Applied AI and Multimedia knowledge graph.

13.3 Interpretation, limitations, and verification

Recommendation Systems is the strongest bridge by inverse-strength weighted betweenness, linking research themes, technical methods, and application outcomes. The graph is synthetic and demonstrates structural interpretation, not empirical evidence about a real research community.

The graph was not extracted from publications. The inverse-strength transformation assumes that stronger semantic relationships represent shorter effective distances. A different substantive meaning for strength would require a different transformation.

Stored node and edge tables, calculated centrality metrics, and static and interactive figures provide reproduction evidence.

14. Discussion

The empirical and synthetic components answer different questions. The Facebook table supports descriptive post comparison but not individual relationship inference. The synthetic graphs reveal how topology, layout, and encoding shape network interpretation.

Mean engagement is influenced by extreme posts, so median summaries, log distributions, and outlier flags accompany it. Correlations remain associational and do not isolate algorithm, audience, or content effects.

15. Limitations and ethical considerations

The data end in 2018, represent ten Thai sellers, and omit reach, impressions, followers, campaign spend, page identity, and content text. Results may not generalize across settings or time.

The project preserves anonymization, reports aggregate findings, avoids re-identification, makes no unsupported causal claims, and clearly labels every dummy graph as synthetic.

16. Conclusion, recommendations, and future work

The largest observed mean engagement was associated with video posts. The supplied small-world graph was presented most clearly by Kamada-Kawai, tie salience was separated from centrality through weighted encoding, and the multidimensional nature of social-network realism was revealed by the model comparison.

For future work, ethically collected reach denominators, multiple seller populations, repeated graph simulations, uncertainty intervals, content representations, and user testing of dashboard accessibility should be added.

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Appendix: Reproducibility and assumptions

The generated artifacts can be rebuilt by running python main.py, and the pipeline can be validated with pytest. The supplied teaching notebooks are source-identical. The official UCI archive was used because Kaggle credentials were unavailable. All assignment relationship graphs are explicitly synthetic.